



We have a csv file to store different electoral surveys for the European elections next June. For each survey we have the following information:

- **Consultant:** Name of the consultant conducting the survey.
- **Start date:** Date of the start of the survey.
- **End date:** End date of the survey.
- **Number of respondents:** Integer with the number of respondents.
- **Country:** Country of the survey.
- **Type of survey:** Means by which the survey is carried out, with values: TELEPHONE, INTERNET, or FACE_TO_FACE.
- **Percentage of undecided:** Percentage of undecided respondents.
- **Party results:** List of pairs with party name and expected vote share.

One line of this file contains the following fields:

```
Consultant B,2024-05-18,2024-05-27,3203,Spain,Internet,16.15,'(Party 1: 42.29);  
(Party 2: 49.52); (Party 3: 6.01); (Party 4: 1.76); (Party 5: 0.18); (Party 6:  
0.23)'
```

This row indicates that Consultant B has carried out a survey that began on 2024-05-18 and ended on 2024-05-27, having surveyed 3203 people. The country where the survey was conducted is Spain, via the internet and with 16.15% undecided. Then, there is a list of the expected vote percentage for six different parties.

Exercise 1: Type Result (0,5 points)

Implement the Result type by means of a record, according to the following information:

Properties:

- **Party**, of type String, observable.
- **Percentage**, of type Double, observable.

Constructors:

- C1: receives one parameter for each basic property of the type, in the same order in which they are defined.

String representation: a string with all the basic properties of the type.

Restrictions:

- R1: The percentage cannot be negative.



Exercise 2: Survey type (1,5 points)

Implement the Survey type using a class, according to the following information:

Properties:

- **Name**, of type String, observable.
- **Start date**, of type LocalDate, observable.
- **End date**, of type LocalDate, observable.
- **Number of respondents**, of type Integer, observable and modifiable.
- **Country**, of type String, observable.
- **Type**, of type SurveyType, observable. It can take the values TELEPHONE, INTERNET, FACE_TO_FACE.
- **Percentage of undecided**, Double type, observable and modifiable.
- **Results**, of type List<Result>, observable.
- **Ratio of respondents per day**, type Double, observable. Ratio of the number of respondents per day of duration of the survey.

Constructors:

- C1: receives one parameter for each basic property of the type.

Restrictions:

- R1: The end date cannot be earlier than the start date.
- R2: The list of results cannot be empty.
- R3: The number of respondents must be greater than or equal to zero.

String representation: a string with all the basic properties of the type.

Equality criterion: two surveys are equal if their start and end date, their consultant, and their number of respondents are equal.

Natural order: two surveys are sorted by start date, by end date, and at the same dates by their consultant, and then by their number of respondents.

Exercise 3: Factory (1 point)

In the class **SurveysFactory**, which is given partially implemented, implement the method:

Survey parseSurvey(String lineCSV): creates a Survey object from a string of characters. The string must have the same format as the lines of the CSV file.

Exercise 4: Sequential Treatments (7 points)

The type SurveysStatistics has the following description:

Properties:

- **surveys**: survey list, of type List<Survey>, observable.

Constructors:

- C1: receives a parameter of type Stream<Survey>.

String representation: a string with all surveys.

Equality criterion: two objects of type SurveysStatistics are equal if their surveys are equal.



Implement the type **SurveysStatistics** and add the following sequential treatments. You must resolve all methods **using streams**, unless you are explicitly told to use loops:

1. *Double getMeanNumRespondentsSurveyConsultantandDate (String consultant, LocalDate dateMaximum)*: Returns the average number of respondents of the surveys carried out by the consultant given as parameter, whose end date is prior to *dateMaximum*. If it cannot be calculated, zero is returned (1 point).
2. *Survey getSurveyMostRespondentsPerDay(SurveyType type)*: Returns the survey of the type given as parameter, that has a highest ratio of number of respondents per day of duration. If unable to calculate, *NoSuchElementException* is raised (1 point).
3. *List<String> getMostFrequentPartiesSorted(Integer n)*: Given a value *n*, returns a list of the *n* parties that appear most often in the surveys, sorted from highest to lowest number of appearances.(1.5 points).
4. *SortedMap<String, Boolean> getSurpassRespondentsByCountry (Integer threshold)*: Returns a *SortedMap*, whose keys are sorted in natural order, where each country is matched by a *Boolean* with a value of *True* if all surveys in that country have a number of respondents higher than the threshold given as parameter (1.75 points).
5. *Map<String, SortedSet<String>> getCountriesPerParty_HigherPercentage (Double thresholdPercentage)*: Given a *thresholdPercentage* of type *Double*, returns a *Map* that relates the name of each party to the set of countries in which it has obtained a percentage of votes higher than the threshold. **Implement this method with loops.** (1.75 points)

Write a **TestSurveysStatistics** class. The class shall read the data from the file and test all sequential treatments, defining a test method for each sequential treatment to be tested. The maximum score of the exercise will not be obtained if the test is not performed and the test runs. The expected results for the dataset provided, with the values indicated in the tests, are:

EJ1 = = = = =

The average of Consultant A with end date before 2024-05-29 is 2993.785714.

The average of Consultant B with an end date prior to 2024-05-26 is 2825.11111111.

EJ2 = = = = =

The TELEPHONE-type survey with the highest ratio of respondents per day is:

Survey [consultant=Consultant B, dateStart=2024-05-06, dateEnd=2024-05-24, numberRespondents=2520, country=Austria, type=TELEFONICA, percentageUndecided=24.56, results=[Result[party=Party 1, percentage=70.41],



```
Result[party=Party 2, percentage=27.99], Result[party=Party 3, percentage=0.57],  
Result[party=Party 4, percentage=0.3]]]]]
```

The FACE_TO_FACE survey with the highest ratio of respondents per day is:

```
Survey [consultant=Consultant C, dateStart=2024-05-11, dateEnd=2024-05-30,  
numberRespondents=2636, country=Germany, type=PRESENTIAL,  
percentageUndecided=36.86, results=[Result[party=Party 1, percentage=30.62],  
Result[party=Party 2, percentage=60.59], Result[party=Party 3, percentage=2.97],  
Result[party=Party 4, percentage=5.37], Result[party=Party 5, percentage=0.28],  
Result[party=Party 6, percentage=0.03], Result[party=Party 7, percentage=0.06],  
Result[party=Party 8, percentage=0.03], Result[party=Party 9, percentage=0.02],  
Result[party=Party 10, percentage=0.01], Result[party=Party 11,  
percentage=0.02]]]]]
```

EJ3 = = = = =

The 5 most frequent parties in the surveys are:

```
[Party 2, Party 1, Party 3, Party 4, Party 5].
```

EJ4 = = = = =

Do all surveys in these countries exceed 3500 respondents?

```
{Austria=false, Austria=false, Belgium=false, Bulgaria=false, Croatia=false,  
Cyprus=false, Denmark=true, Estonia=false, Finland=false, France=false,  
Germany=false, Greece=false, Hungary=false, Ireland=false, Italy=false,  
Latvia=false, Lithuania=false, Luxembourg=false, Malta=false, Netherlands=false,  
Portugal=false, Slovenia=false, Slovakia=false, Slovenia=false, Spain=false,  
Greece=false, Hungary=false, Ireland=false, Italy=false, Latvia=false,  
Lithuania=false, Luxembourg=false, Malta=false, Netherlands=false, Poland=false,  
Portugal=false, Romania=false, Slovakia=false, Slovenia=false, Spain=false,  
Sweden=false}
```

EJ5 = = = = =

Countries where each party has won more than 90% of the vote:

```
{Party 2=[France], Party 1=[Austria, France, Germany, Greece, Ireland, Italy,  
Latvia, Lithuania, Netherlands, Portugal, Romania, Spain, Sweden, Sweden]}
```

Countries where each party has won more than 97% of the vote:

```
{Party 1=[France, Greece, Latvia, Lithuania]}
```